

tf.compat.v1.nn.conv2d Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/conv2d

Computes a 2-D convolution given 4-D input and filter tensors.

Function Signatures

```
tf.compat.v1.nn.conv2d( input, filter=None, strides=None,
padding=None, use_cudnn_on_gpu=True, data_format='NHWC',
dilations=[1, 1, 1, 1], name=None, filters=None )
```

Code Examples

```
tf.compat.v1.nn.conv2d(  
    input,  
    filter=None,  
    strides=None,  
    padding=None,  
    use_cudnn_on_gpu=True,  
    data_format='NHWC',  
    dilations=[1, 1, 1, 1],  
    name=None,  
    filters=None  
)
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    name=None,  
    filters=None  
)
```

```
output[b, i, j, k] =  
    sum_{di, dj, q} input[b, strides[1] * i + di, strides[2] * j + dj,  
    q]  
    * filter[di, dj, q, k]
```

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output[b, i, j, k] =  
    sum_{di, dj, q} input[b, strides[1] * i + di, strides[2] * j + dj,  
    q]  
    * filter[di, dj, q, k]
```

Documentation

Computes a 2-D convolution given 4-D input and filter tensors.

Given an input tensor of shape `[batch, in_height, in_width, in_channels]` and a filter / kernel tensor of shape `[filter_height, filter_width, in_channels, out_channels]`, this op performs the following:

- Flattens the filter to a 2-D matrix with shape

`[filter_height * filter_width * in_channels, output_channels]`.

- Extracts image patches from the input tensor to form a virtual

tensor of shape `[batch, out_height, out_width, filter_height * filter_width * in_channels]`.

- For each patch, right-multiplies the filter matrix and the image patch

vector.

In detail, with the default NHWC format,

Must have `strides[0] = strides[3] = 1`. For the most common case of the same horizontal and vertical strides, `strides = [1, stride, stride, 1]`.

Returns

